RAJALAKSHMI
ENGINEERING
COLLEGE

Thamizhiniyan P 251001284-2025-IT T2

Started on	Friday, 11 September 2026, 10:11 PM
State	Finished
Completed on	Friday, 11 September 2026, 10:13 PM
Time taken	1 min 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- 1 <= n <= 5 * 10^4
- -2^{31} <= nums[i] <= 2^{31} - 1

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int count(int arr[], int left, int right, int value)
{
    int c = 0;
    for (int i = left; i <= right; i++)
    {
        if (arr[i] == value)
            c++;
    }
    return c;
}

int majorityElement(int arr[], int left, int right)
{
    if (left == right)
        return arr[left];

    int mid = (left + right) / 2;

    int leftMajority = majorityElement(arr, left, mid);
    int rightMajority = majorityElement(arr, mid + 1, right);

    if (leftMajority == rightMajority)
        return leftMajority;

    int leftCount = count(arr, left, right, leftMajority);
    int rightCount = count(arr, left, right, rightMajority);

    if (leftCount > rightCount)
        return leftMajority;
    else
        return rightMajority;
}

int main()
{
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    printf("%d\n", majorityElement(nums, 0, n - 1));

    return 0;
}
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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